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1.1 Abstract

Fake news now ranks among today's toughest digital problems - its speed on social networks threatens elections, health decisions, even how communities hold together. Because human reviewers can't keep up with the sheer flood of false claims, tools that spot deception without help are gaining ground.

This work builds and tests a system that spots false claims in text, relying on machine learning along with transformer methods, tested against the LIAR collection - a set of 12,836 brief political remarks pulled from PolitiFact.com. Four classical models are evaluated as baselines — Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine, and Random Forest, each leaning on TF-IDF to pull out word patterns. Then came DistilBERT, adjusted in one run using just wording, yet in another blending extra details like past dishonesty rates and political ties. Performance shifted when background facts joined the words.

Accuracy figures sit around 60–62% for traditional models, Logistic Regression edging ahead with 62.59%, alongside a macro F1 score of 0.607. Though Text-only DistilBERT hits similar accuracy levels, its macro F1 dips slightly to 0.590 - this aligns with earlier reports indicating limited gains from transformers when handling brief statements like those in the LIAR dataset. When metadata gets woven into DistilBERT, performance climbs: 65.51% accuracy appears, along with a macro F1 reaching 0.645. That setup outperforms all five others tested. Speaker-related background details clearly contribute useful distinctions, going further than words by themselves.

Using LIME and SHAP, researchers examined how Logistic Regression and DistilBERT with Metadata make decisions - findings show words comparing quantities strongly influence false outcome labels in both systems. Through a live demo built with Gradio, each of the five trained setups classifies inputs instantly while showing supporting evidence via LIME visuals.

1.2 Introduction

The dramatic increase in use of social media and digital news sites has dramatically changed how people consume and share information. As with many good things about our increasing reliance on digital media; however, the same forces have provided an opportunity for the dissemination of false information at rates previously unheard of. "Fake News" is a term used to describe intentionally fabricated or misleading information (or even false images) made to appear like actual news reports. Fake News is a threat to democracy, public health, and social cohesiveness. In addition to high profile events such as the misinformation surrounding health issues caused by the COVID-19 virus and the civil unrest caused when false news was reported regarding the situation in Southport in 2024 show us that misinformation is harmful beyond just being a problem in the digital space.

Manual organizations, which include PolitiFact, Snopes, and Full Fact, are working hard to counteract the effects of disinformation. However, these organizations' manual workflows will never be able to operate at the scale needed to combat the sheer number of pieces of information that are generated every day through various social media platforms. Therefore, there is a clear need for automated solutions capable of detecting fake news in near-real-time.

This project developed a system to detect fake news based on text input utilizing Machine Learning and Natural Language Processing techniques. The LIAR Dataset, one of the most established benchmarks available, was utilized for evaluation purposes. The LIAR Dataset contains over 12,000 very short statements labeled for truthfulness/veracity. The four classical machine learning models were included as baseline models: Logistic Regression, Naive Bayes, Linear Support Vector Machine (SVM), and Random Forest. These models were compared to a fine-tuned version of the DistilBERT Transformer Model. An additional configuration of the transformer model includes speaker metadata from the LIAR Dataset, specifically: lie count history and party affiliation. The question posed here is if context outside of text will improve model accuracy. LIME and SHAP explainability methods were employed to provide transparent reasoning for each model's predictions, providing answers to the black box nature of machine learning models. A Gradio-based

web interface is developed as a feasibility demonstration, allowing users to input statements, select models, and receive real-time classifications with explanations.

The intended users of this system are journalists, fact-checkers, and members of the general public who wish to verify the credibility of news statements. The system is designed with transparency and accessibility in mind, recognising that explainability is as important as accuracy for real-world adoption.

1.3 Background Research

1.3.1 The Fake News Problem

Fake news refers to deliberately fabricated or misleading information presented as legitimate news content, designed to deceive readers for political, financial, or social purposes. The digital age has dramatically accelerated how such misinformation spreads, posing significant threats to information ecosystems (Singh et al., 2026). Unlike honest errors or clearly marked satire, fake news is intentionally created to manipulate public opinion, influence electoral outcomes, and exploit cognitive biases.

The real-world consequences of fake news are severe. During the 2016 U.S. presidential election, fabricated content achieved wider distribution on social media than reporting from established news organisations (Ahmed et al., 2017). In July 2024, false information about a tragedy in Southport led to widespread civil disturbances in the UK. During the COVID-19 pandemic, false claims about treatments and vaccines contributed to preventable deaths. These examples demonstrate that fake news is not merely an information quality problem but a public safety issue.

Manual factchecking is fundamentally inadequate at scale. Misinformation spreads six times faster than truthful information on digital platforms, and approximately 68% of Americans now obtain news from social media (Lee et al., 2023). Organisations such as PolitiFact and Snopes provide valuable services but cannot match the velocity of misinformation propagation. Analysis of suspicious Telegram accounts found coordinated disinformation campaigns involving over one million messages from more than 2,000 accounts, operating at a scale that makes manual review impossible. This demonstrates the necessity of automated detection.

1.3.2 Traditional Machine Learning Approaches

Older methods of fake news detection were based on classic machine learning models such as Naive Bayes, Support Vector machines, Decision trees, and Random Forests (Kaliyar et al., 2021). Such approaches derive human-crafted characteristics including word counts and n-grams and use them to train statistical models to differentiate real and fake news. According to studies, the accuracy of the algorithms varies between 74% and 96% based on the algorithm and dataset (Granik and Mesyura, 2017; Reddy et al., 2019).

SVMs have been found to be especially useful in identifying the best decision boundaries in high-dimensional TF-IDF feature spaces. Ensemble learning lessens overfitting in random Forest classifiers. Naive Bayes provides relatively high computational efficiency, and bigram-based models can be as accurate as 93% on mixed datasets (Kaliyar et al., 2020), but the feature independence assumption often acts as a bottleneck to performance relative to SVM and Random Forest.

Nonetheless, conventional approaches have inherent weaknesses as per these findings. They are unable to comprehend context and subtle language sarcasm, metaphor, and specialized phrasing are hard to discern by surface-level characteristics alone (Kaliyar et al., 2021). There is a similar issue of domain shift: models trained on political news do not generalise to health or entertainment (Qian et al., 2024). These limitations motivate the use of transformer-based models.

1.3.3 Transformer-Based Models and Modern NLP

Transformer-based LMs like BERT, RoBERTa, and ALBERT are bidirectionally trained on text with attention mechanisms and are sensitive to semantic and long-range dependencies, which other models do not capture

(Kaliyar et al., 2021). This renders them especially useful in dealing with such complex linguistic phenomena like sarcasm and contextual meaning that are often found in misinformation.

Empirical comparisons show that there are significant performance differences. Kaliyar et al. (2021) trained FakeBERT, a hybrid of BERT and convolutional blocks with 98.90% accuracy. RoBERTa has demonstrated 89.23% precision and 90.14% recall when performing benchmark tasks, whereas ALBERT has surpassed other variants with 87.6% accuracy and time-efficient training (Azizah et al., 2023). These performance results, however, are mostly on datasets not LIAR - all models show significantly lower performance on LIAR because of its short average statement length and ambiguity in labels.

Transformer models pose serious practical difficulties. In training, BERT has about 110 million parameters and scales to $O(L^2 \times d)$ which is resource-intensive. To solve this, DistilBERT is able to reduce the size of its model by 85 percent but maintain the same performance of BERT, which has inspired the use of the model in this project.

1.3.4 Benchmark Datasets for Fake News Detection

The LIAR dataset (Wang, 2017) consists of 12836 human-rated short statements found on PolitiFact.com and rated with one of six veracity labels: pants-fire, false, barely-true, half-true, mostly-true and true. The average statement length is 19 words which makes the dataset appropriate in assessing detection of short social media posts. In binary classification, the six labels are grouped into two categories (Fake (pants-fire, false, barely-true) and Real (half-true, mostly-true, true)) according to the methods of a previous study (Galli et al., 2021). Most models with sophisticated transformer architecture can be configured to reach just 59 to 65 percent accuracy on LIAR, despite standard settings, which is indicative of the challenge of identifying subtle misinformation in brief claims.

Other datasets like ISOT and FakeNewsNet were not chosen as well as Liar due to a number of reasons. All actual articles in ISOT were gathered on Reuters and start with the term Reuters so that models could learn the publishing patterns instead of actual content properties (Iceland, 2024). FakeNewsNet has longer articles with an average of 1,459 words per article, which is not as appropriate in assessing short-text political claims. LIAR circumvents these biases, and also incorporates metadata about the speaker, such as the number of lies and party affiliation, which offers additional signal on top of the text content.

1.3.5 Explainability in AI-Based Fake News Detection

The way models make their decisions is a key aspect of establishing trust and making ethical use in journalism and fact-checking (Singh et al., 2026). State-of-the-art NLP models are commonly black boxes, meaning they make predictions without clear explanations, thus decreasing user trust and reducing adoption (Nwaiwu et al., 2025).

LIME interprets individual predictions by locally approximating the classifier with a linear model, which is computationally efficient and can be interpreted on the individual prediction level (Dua et al., 2023). SHAP offers both global and local explanations based on game-theoretic Shapley values and is capable of identifying non-linear relationships that LIME might fail to capture as a result of its linear approximation (Mezzi et al., 2024). Human assessment experiments prove the usefulness of these techniques - 82% of the explanations produced by hybrid LIME and SHAP models were considered meaningful by the users, which proves that transparency does boost trust (Nwaiwu et al., 2025).

1.3.6 Existing Fake News Detection Products and Platforms

The fake news detection landscape consists of both hand-operated fact-checking organisations and new AI-based tools that have different capabilities and limitations.

PolitiFact was started in 2007 and has a six-point Truth-O-Meter scale, which was awarded the Pulitzer Prize in 2009 (Lee et al., 2023). Snopes is the best-known and oldest fact-checking site, having been established in 1994 and with a five-point rating scale with other categories such as Outdated and Satire. Other nonpartisan verification services are offered by Full Fact and FactCheck.org. A study between 24,000 Snopes and PolitiFact fact-checking articles published in 2016-2022, in turn, discovered a concordance rate of high

accuracy, yet a single contradictory rating was identified among hundreds of identical claims (Lee et al., 2023). Although they are reliable, manual platforms are not as fast as misinformation spreads.

Despite these developments, there are still critical gaps. The vast amounts of misinformation in the contemporary world cannot be dealt with by most manual platforms. Most AI tools specialize in a certain modality or platform as opposed to being comprehensive in detection. The vast majority of them are reactive, there is little enough explainability to give them the confidence of the user, and many commercial tools are not accessible by individual users. Such restrictions encourage the creation of a system that is both accurate, transparent, and accessible.

1.3.7 Professional, Legal, Ethical, and Social Issues (PLESI)

Automated fake news detection systems raise profound questions across professional, legal, ethical, and social dimensions that must be carefully considered in system design.

Ethical Issues. The inherent ethical conflict of automated fake news detection is content moderation versus censorship. Whether to flag certain content will always come with value judgements as to what reasonable people will disagree. Any system that only focuses on statements that are literally false can fail to detect deliberately deceptive content, whereas a more liberal set of standards can be accused of politically-minded censorship. The algorithmic bias is another issue - the experiences and biases of human annotators are directly copied to trained models and may cause disproportionate labeling of content based on a specific political view or group of people. The question of whether harmful disinformation is to be distinguished or not, versus satire, opinion, or justified political commentary is another difficulty, because excessive aggressive detection may lead to silencing speech that ought not to be silenced.

Legal and Regulatory Frameworks. Rules differ across regions. In Europe, the GDPR restricts targeting people using their political opinions while giving individuals control over decisions made by algorithms - this complicates AI systems that rely on mass data collection. Platforms reaching 45 million users monthly must follow the DSA's rules to manage false information; failing means paying fines as high as 6 percent of worldwide earnings. Across the channel, Britain's Online Safety Act demands evaluations of risks tied to fake videos and artificial content, backed by penalties hitting £18 million or one-tenth of total income. Meanwhile, America lacks unified national laws, depending instead on varying state policies, leaving international operations in doubt.

Social Implications. Around 38 percent of content posted on Facebook by conservative political websites contains inaccurate or deceptive claims, studies show. Because individuals tend to embrace details matching their prior beliefs, such patterns fuel both the circulation of falsehoods and scepticism toward verification attempts. When algorithms judge what counts as true, worries emerge over free speech erosion. Such tools might concentrate interpretive power, forming dominance over narrative control - yet unmonitored deception also erodes public judgment. Design and rollout phases need constant attention to these tensions, balancing accuracy against openness. One misstep tilts the scale too far either way.

1.3.8 Synthesis and Restated Project Aims

The literature review has shown that there are several critical insights that will directly inform the design of this project. The classic machine learning methods obtain reasonable accuracy in the text classification task but have a fundamental limitation in that they are sensitive to surface-level features only - they are unable to deal with context, sarcasm or domain shift, and their performance drops dramatically when used out of context. Transformer-based models, especially BERT and its variations, significantly outperform traditional methods on most of the benchmarks, with a 95-99% accuracy in controlled contexts. Nonetheless, these findings are mostly on datasets including ISOT which includes publisher-identifying artefacts, which inflates reported performance. Even the best transformer architectures have only 59-65% accuracy on truly challenging benchmarks such as LIAR, which indicates that state-of-the-art transformer architectures do not yet support the verification of political claims on short text.

One of the main lessons in the literature is that the quality of the dataset and the availability of features are as important as the model structure. The near-perfect transformer results of the ISOT dataset fail once patterns

of publishers are eliminated, showing that high benchmarks accuracy does not imply applicability in practice. This result is what directly prompts the decision to use LIAR as the main benchmark and explore the concept of metadata of speakers as an additional signal in case the text alone cannot be used as the discriminative signal, the lack of which is compensated by the presence of contextual source-level information about the credibility of the speakers.

Explainability becomes a need of practical deployment. Research indicates a consistent pattern, regardless of accuracy, where black-box classifiers are not adopted in journalism and factchecking since users are unable to determine the rationale behind predictions. LIME and SHAP are complementary and SHAP is more comprehensive in global explanations, whereas LIME offers fast, local explanations, and human evaluation research studies show that hybrid explanation methods are perceived meaningful by most users.

Current detection platforms demonstrate high gaps: manual platforms are not scalable, AI tools are not generalisable and explainable, and most cannot be used by general users. These loopholes are the direct reasons behind the creation of a system that embraces a combination of different types of models, explainability in a transparent manner, and an simple interface.

According to these results, this project will: apply and compare four classical machine learning baselines with DistilBERT to the LIAR dataset; explore whether speaker metadata can be used to enhance transformer model performance on short-text classification; apply LIME and SHAP to understand model predictions in a transparent way; and create an interface based on Gradio to provide real-time classification with explainability output. It aims at having a system that trades accuracy, efficiency, explainability and ethical responsibility in a unified deployable system.

1.4 Methodology

1.4.1 Research Approach

The empirical and experimental research approach used in this project is in line with the methods used in other similar fake news detection research (Kaliyar et al., 2021; Azizah et al., 2023). The methodology does not just follow theoretical analysis but entails the building of working models, controlled experiments, and conclusions based on quantifiable outcomes.

The project is based on the CRISP-DM model, a popular process model of data science projects, which outlines iterative steps of data cognition, data preparation, modelling, evaluation, and implementation (Wirth and Hipp, 2000). This framework is very appropriate to the project since each step is informed by the previous step- results of an evaluation directly informed the intention to extend the original DistilBERT framework with metadata features. CRISP-DM was iterative, and it was possible to refine the stage instead of making any promises beforehand to a fixed pipeline.

The software artefact was developed using an agile approach. All the components, such as baseline models, transformer model, explainability module and interface were developed and tested separately and then integrated. This minimized risk and provided issues to be detected at the initial stages and not when the entire system was assembled.

1.4.2 Dataset Selection

The choice of data here centers on the LIAR dataset (Wang, 2017), serving as the main foundation. Sourced from PolitiFact.com, it holds 12,836 brief claims reviewed by people, each tagged with one of six truth levels: pants-fire, false, barely-true, half-true, mostly-true, or true. When sorting into just two groups - Fake and Real - the first combines pants-fire, false, and barely-true; the second includes half-true, mostly-true, and true. This split mirrors earlier research practices seen in Galli et al. (2021).

One reason LIAR stands out lies in its role as a trusted reference point within research circles. Because earlier studies have relied on it, drawing parallels becomes straightforward when evaluating new findings. Unlike ISOT, where hidden markers tied to sources can mislead algorithms into focusing on irrelevant details instead of actual language patterns - a flaw noted recently - this collection avoids such pitfalls. Its difficulty level stays

high, so even sophisticated systems show noticeable variation in outcomes. Within the structure, examples get divided: most go toward learning patterns, a smaller portion helps tune settings, while another set remains untouched until final checks. Every model faces identical splits during trials, keeping evaluations balanced from start to finish.

1.4.3 Preprocessing Pipeline

The same text preprocessing pipeline was used to all four baseline models in order to make sure that the variation in performance of the models is due to the models themselves and not variation in input data. The pipeline carries out the following sequentially: all the text is turned to lower-case, URLs are removed, special characters and punctuations are removed, and all the whitespace is regularised into single spaces. Removal of stop words was also a possibility but was deliberately left out because stop words like end of and more than n-gram features provide discriminative information in the case of political text classification.

All baseline models have their features extracted using TF-IDF vectorisation with the top 5,000 features (maximum feature count), unigrams and bigrams (ngram_range = 1,2), minimum document frequency of 2, and maximum document frequency of 0.8. The vectoriser was only applied to training data without re-fitting and the same applied to validation and test sets to avoid data leakage. This methodology is in line with conventional practice in text classification studies (Kaliyar et al., 2020).

In the case of the DistilBERT models, raw text was directly fed to the HuggingFace tokeniser and a maximum sequence length of 128 tokens was used. No pre-processing other than whitespace normalisation was performed as there is no purpose of the transformer model running on raw text since extra cleaning can remove contextual information that the model will use.

1.4.4 Baseline Model Selection

A baseline setup includes four traditional machine learning methods: Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine (SVM), alongside Random Forest. Selection of these models stems from their prevalence in foundational studies on fake news identification (Kaliyar et al., 2020; Granik & Mesyura, 2017; Shetty et al., 2019). Each brings a distinct way of reaching decisions, enabling broad comparative analysis across varied algorithmic behaviours.

Starting with Logistic Regression - it serves as the main reference point because it balances clarity with solid results when dealing with cleanly divided text data. Although Multinomial Naive Bayes brings speed and a probability-driven approach, its reliance on independent features often hampers effectiveness in nuanced language contexts. Linear SVM stands out by maximizing separation margins, making it effective even when handling sparse vectors typical of TF-IDF representations. Random Forest counters model instability by combining many individual trees, while also delivering insights via Gini importance metrics.

Each of the four models undergoes training and assessment under matching conditions - same data preparation, transformation into vectors, and testing methods - to allow balanced comparisons. Instead of adjusting settings, default values stay unchanged across runs: Logistic Regression uses 1000 maximum iterations; Naive Bayes applies a smoothing value of 1.0; SVM sets penalty strength at 1.0 and iteration limit at 2000; Random Forest builds 200 trees while selecting features based on square root rules. These choices aim not to push peak results but to create consistent starting points. Fairness matters more than optimisation here.

1.4.5 Transformer Model Selection

Chosen for this work, DistilBERT (Sanh et al., 2019) serves as the foundation among transformer architectures. Though smaller, it keeps nearly 97% of BERT's grasp on language structure - despite being cut down in size by two-fifths. Speed gains during prediction rise sharply, hitting around 60% faster processing. Because of these traits, running experiments becomes more feasible under limited computing power. At the same time, results can still line up against studies using complete BERT variants found across prior research.

A classification head gets attached to the pre-trained DistilBERT during fine-tuning, aimed at binary sequence labeling via HuggingFace Transformers. Training runs across five passes through the LIAR dataset, adjusting

weights with a 2e-5 learning rate while processing 16 examples per batch. Although larger variants exist, DistilBERT fits better under tight compute limits without sacrificing much accuracy. Earlier work by Devlin et al. (2019) supports these settings when adapting transformer models to specific classification jobs. Short statements in datasets like LIAR respond well even to lighter architectures - this helps explain why efficiency wins here.

1.4.6 Evaluation Strategy

Each model undergoes assessment through an identical group of measures: accuracy, precision, recall, per-class and macro F1-scores, along with confusion matrices. Relying only on accuracy proves misleading because of uneven class distribution in the LIAR test data - 714 Real cases compared to 553 Fake ones - a bias that lets a simplistic model labeling everything as Real hit 56% correctness while gaining no real insight. Because it combines precision and recall into one balanced view, the F1-score takes priority when comparing performance across systems.

Looking at mistakes helps check how well each base model performs. Twenty wrongly classified cases from every model get sorted by hand, spotting repeated issues in their output. Because of this close review, we see what numbers alone cannot show - that basic language traits often miss deeper flaws when catching false stories.

Though training progresses, the validation data checks accuracy for DistilBERT. Performance numbers shown later come only from the unseen test collection.

1.4.7 Explainability Methods

In order to deal with the black-box nature of machine learning models, specifically DistilBERT, the LIME and SHAP were introduced to the project. LIME interprets individual predictions by approximating the classifier locally by a linear model, being both computational and interpretable at the per-prediction level (Dua et al., 2023). SHAP can give both global and local explanations with game-theoretic Shapley values and can find non-linear relationships that LIME can overlook because it uses a linear approximation (Mezzi et al., 2024).

Both of these approaches were used to Logistic Regression and DistilBERT + Metadata. When explainability is applied to both a classical and a transformer model, it is possible to directly compare the way the different architectures come to their decisions, and determine whether the two model types are using similar or different linguistic signals.

1.4.8 Interface Development

A simple web-based interface was created as a feasibility prototype based on Gradio, a Python library to quickly prototype machine learning applications. Gradio was selected instead of more complicated frameworks like Flask or Django, and its simplicity suits a proof-of-concept tool and can be directly integrated with HuggingFace models without the need to have its own backend server.

The interface gives users the choice of five available models, to input a news statement and to get a real-time prediction with a confidence score and a LIME explanation chart. The system is not meant to be deployed in production but will show that the trained models can be made available in a usable, interactive format, fulfilling the software artefact criterion of the project brief.

1.4.9 Tools and Programming Languages

The main programming language of the project was Python 3.10 because of its access to all the major machine learning and NLP libraries. Baseline model implementation and TF-IDF vectorisation were implemented using scikit-learn. The pre-trained HuggingFace Transformers DistilBERT model and tokeniser were available together with PyTorch as the deep learning back-end to fine-tune and custom model architecture. The explainability analysis was performed by LIME and SHAP, and Gradio was utilized to create the interface. Pandas and NumPy were used to perform data manipulation and preprocessing, matplotlib and seaborn were used to visualise the results. Jupyter Notebooks were developed to achieve development through a single notebook that was kept per stage of modelling to maintain clarity and usability.

1.5 Implementation and Findings

1.5.1 Dataset Preparation and Preprocessing

LIAR data were loaded in its original TSV format where there were 10,240 training, 1,284 validation, and 1,267 test samples with 14 columns such as statement text, veracity label, name of the speaker, job title, party affiliation, and number of past lies. The six initial veracity labels were merged into a binary classification system: statements labeled pants-fire or false or barely-true were categorized in Fake class (0), and half-true, mostly-true and true statements were categorized in Real class (1). This binarisation is based on that of previous studies (Galli et al., 2021) and gave a slightly skewed test sample of 553 Fake and 714 Real samples. The primary comparative metric was thus macro-averaged F1 instead of the accuracy alone.

In the case of the baseline models, the preprocessing pipeline in Section 1.4.3 was uniformly used in all the four models. TF-IDF vectorisation resulted in a 5,000 feature matrix that was trained on training data only. In the case of the DistilBERT models, raw statements were sent to the HuggingFace tokeniser with a sequence maximum length of 128 tokens - enough since the average LIAR statement length is 19 words.

1.5.2 Baseline Model Results

Each of the four baseline models went through the same preprocessing and vectorisation steps during training and evaluation, so any variation in performance can be attributed to the models, not data handling. The complete outcomes are shown in Table 1.

Model	Accuracy	Fake Precision	Fake Recall	Fake F1	Real F1	Macro F1
Logistic Reg.	62.59%	0.590	0.468	0.522	0.693	0.607
Naive Bayes	62.35%	0.592	0.443	0.507	0.696	0.601
Linear SVM	60.38%	0.548	0.530	0.539	0.653	0.596
Random Forest	62.19%	0.595	0.420	0.492	0.699	0.596

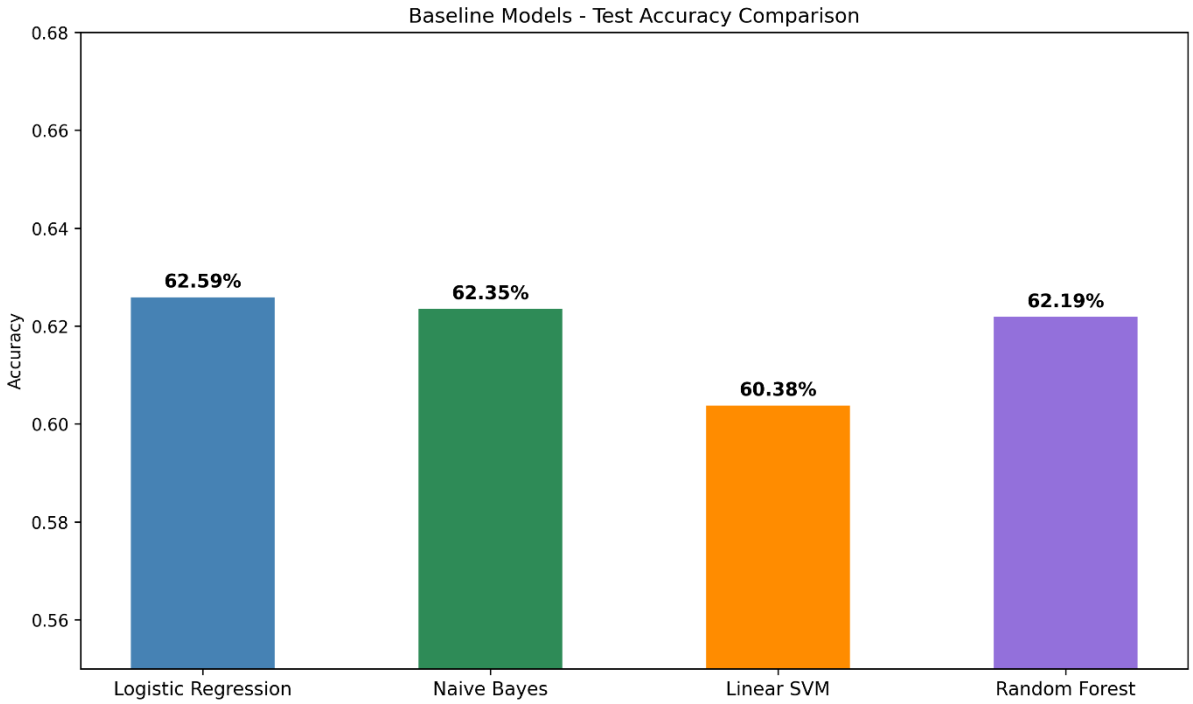


Figure 1: Baseline model test accuracy comparison

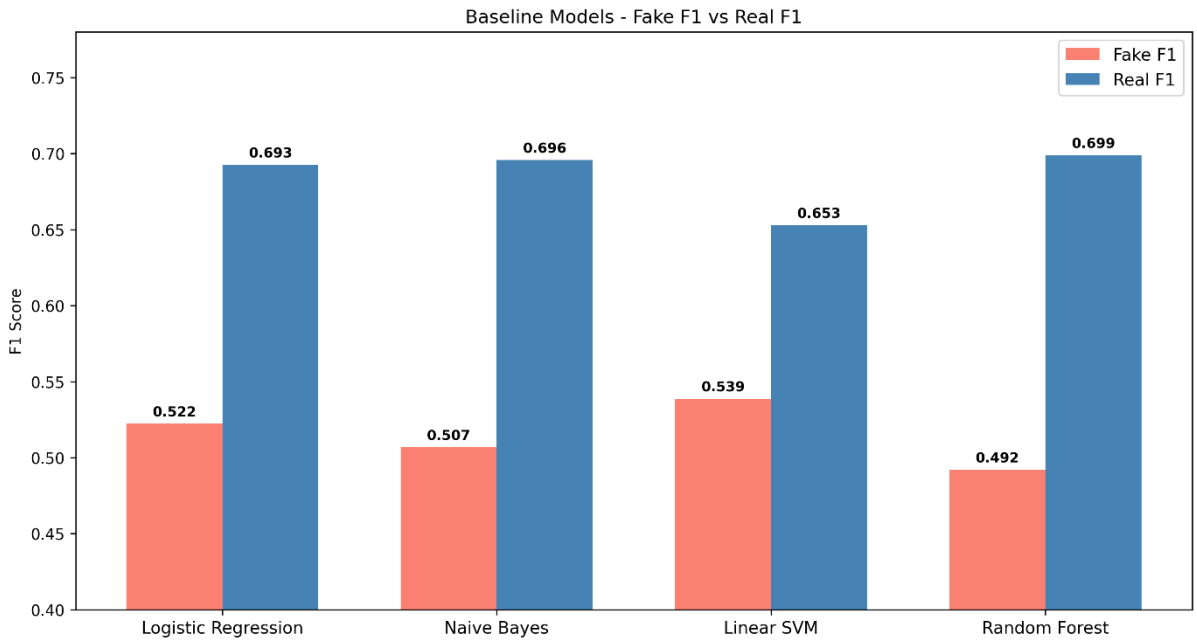


Figure 2: Fake F1 vs Real F1 comparison across baseline models

Starting off, Logistic Regression reached the top accuracy mark - 62.59% - alongside a leading macro F1 score of 0.607, emerging as the most effective baseline method here. That kind of result lines up with earlier findings: logistic regression tends to handle high-dimensional TF-IDF data efficiently because it draws clear linear separations even when text inputs are sparse (Kaliyar et al., 2020). Naive Bayes came close in accuracy, hitting 62.35%, though its macro F1 dipped slightly to 0.601, which hints at limitations tied to assuming all features act independently. Interestingly, Random Forest landed at 0.596 in macro F1, just level with Naive Bayes, yet required 200 combined decision trees - an increase in processing demands without much improvement on this task. At the bottom for accuracy was Linear SVM with 60.38%, however it stood out in another way: achieving the highest Fake class F1 at 0.539, showing its error pattern spreads more evenly across both labels compared to others.

One clear trend appears in all four models: Real F1 scores stay well above Fake F1, differing by at least 0.153 (SVM) and up to 0.207 (Random Forest). Such a difference stems from two causes. A slight imbalance in the test data exposes each model to slightly more genuine samples during learning. Beyond that, true statements in LIAR show sharper linguistic signals - political facts often include official terms, concrete names, and detailed references, patterns that TF-IDF can detect effectively. In contrast, false entries, especially those marked barely-true, mimic truthful ones closely, making them tougher to distinguish using only shallow traits.

1.5.3 Feature Importance Analysis

Looking at feature importance reveals what language cues matter most to each model. In Logistic Regression, terms like "since", "than", "countries", "day", "percent", and "more" strongly signal the Fake category - these often appear in comparisons or numerical assertions. On the other hand, words such as "wisconsin", "obama", "obamacare", and "rep" are linked to Real labels, pointing toward concrete names and political figures. What stands out is how general comparison markers tie more closely to false claims, whereas precise real-world references tend to align with factual reports from PolitiFact. One sees a clear distinction: abstract qualifiers drift toward fabrication, yet particular entities anchor truth.

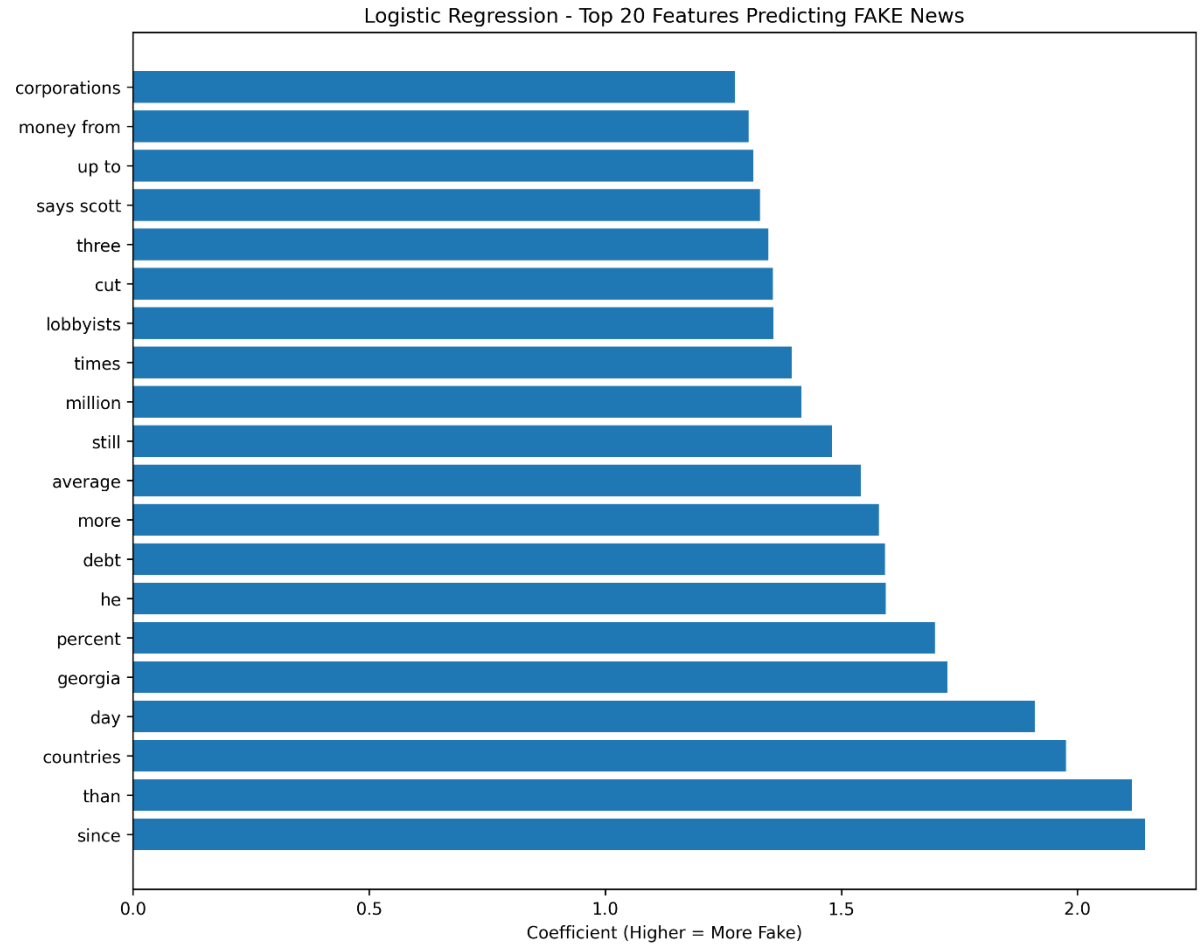


Figure 3: Logistic Regression — Top 20 features predicting Fake news

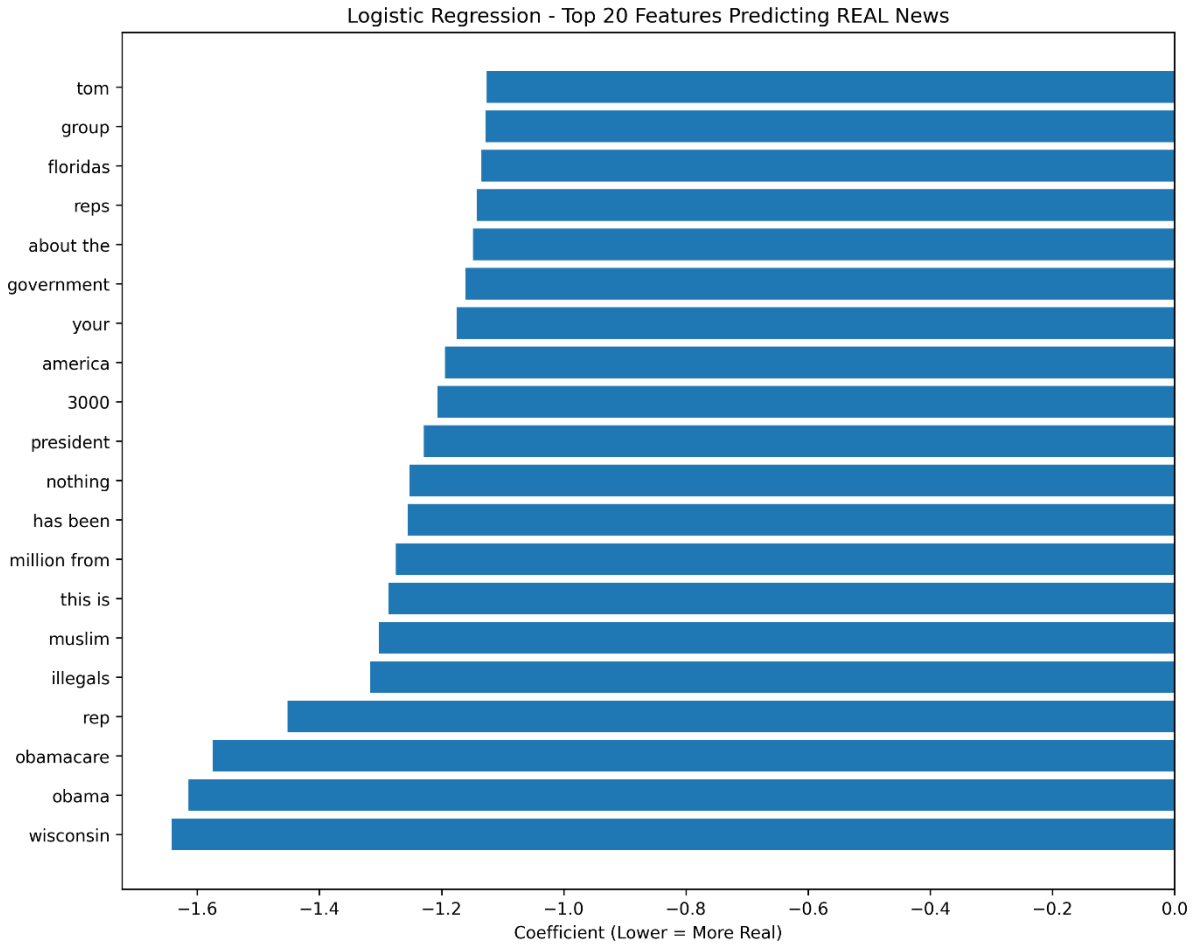


Figure 4: Logistic Regression — Top 20 features predicting Real news

Unexpectedly, log probabilities in Naive Bayes show recurring signals - phrases like "government takeover", "deciding", "socialists" lean toward Fake, whereas terms including "accidents", "incarceration", "graduation" point more to Real. Feature relevance in Random Forest, measured by Gini scores, singles out "percent", "million", "obama", "since" as top differentiators across data, aligning closely with what logistic regression detected.

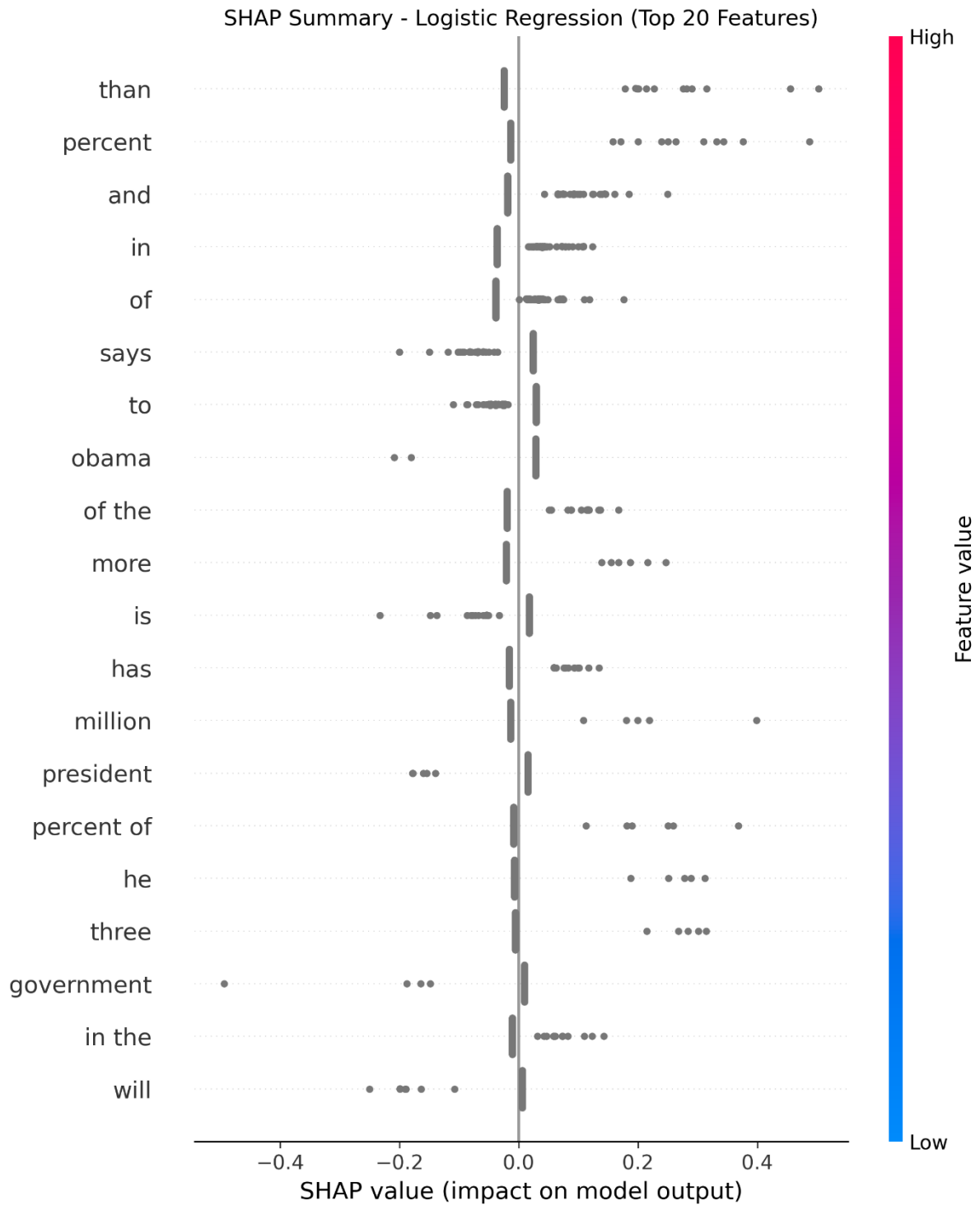


Figure 5: SHAP summary plot — Logistic Regression (top 20 features across 100 test samples)

1.5.4 Error Analysis

A close look at mistakes began with each of the four starting models along with the text-based DistilBERT version, where two dozen wrong predictions got sorted into groups for every one. One label applied when meaning stayed unclear because wording gave too little to go on. Another covered cases where numbers lent deceptive weight to untrue statements. Some errors showed correct facts twisted by slanted presentation. Subjective views mistaken for objective truth formed a separate cluster. The last group caught falsehoods that simply took odd forms not fitting elsewhere.

Every model shows a similar trend: about half to three-fifths of mistakes fall into the unclear category. Following behind, counting roughly one out of every four errors, come those tied to numbers. Despite differences between systems, their weak spots often match up. Certain examples tripped up each version without exception. One example was a claim discussing immigration figures using broad estimates - phrased

more like doubt than deception - which machines wrongly flagged. Another stumbling block appeared in minimal phrases focused only on subjects, such as a fragment mentioning energy policy with nearly nothing else to go on.

This analysis points toward a key issue: it is less about which model was picked and more about how tough the LIAR dataset really is. Quite a few errors come from cases demanding background insight or situational awareness - things no text-only system can grasp just by reading words. Because of this, looking into metadata within transformer frameworks becomes a natural next step.

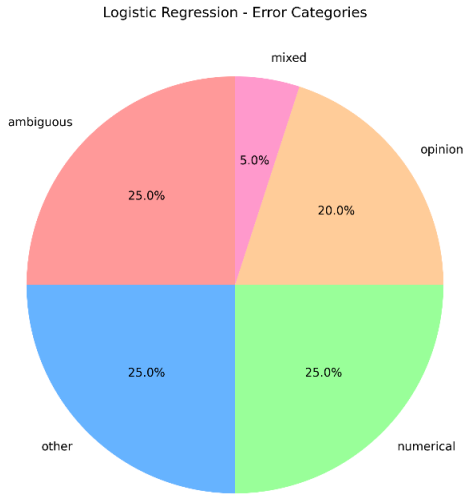


Figure 6: Logistic Regression Error category distribution

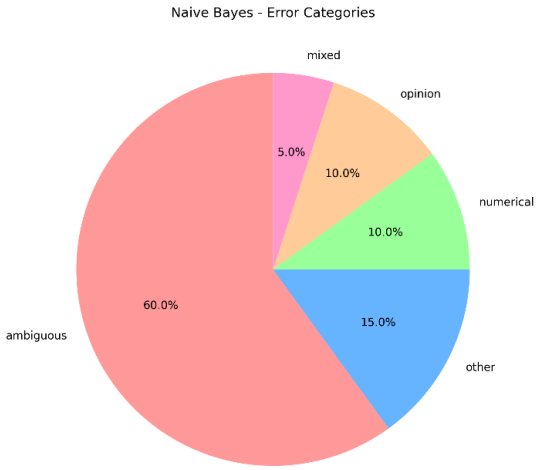


Figure 7: Naive Bayes Error category distribution

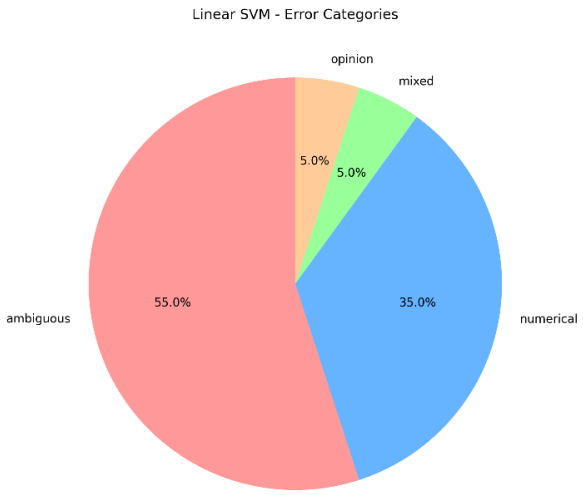


Figure 8: Linear SVM Error category distribution

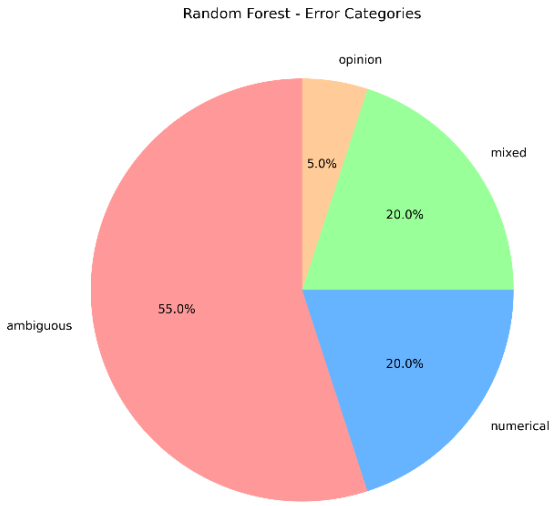


Figure 9: Random Forest Error category distribution

1.5.5 Transformer Model Results

A different version of DistilBERT was tested alongside another setup. One used only textual data, adjusted through training across five cycles with a step size of 0.00002 and processed sixteen samples at once, using a modified error measure to handle uneven category representation. It reached an evaluation score of 62.59 percent correct answers and scored 0.590 on average for precision and recall combined - matching logistic regression's correctness yet falling short in balanced performance metrics. Not managing to beat a straight-line predictor when relying solely on words marks a key observation. Transformer models are particularly effective at capturing long-range dependencies and contextual meaning in long text, but it becomes much less effective on short inputs. The mean LIAR statements of only 19 words provide a lack of context in which the bidirectional attention mechanisms can benefit, which is why the use of text-only DistilBERT does not provide an advantage over Logistic Regression on the dataset.

Instead of relying on text alone, the updated setup added background details about speakers. For every claim in the LIAR set, it logs how often the speaker received ratings like pants-fire or mostly-true before, plus their political alignment. Past trustworthiness shows up here - something the words themselves do not reveal. To process both types of input, one method linked the output from DistilBERT's [CLS] position with scaled numerical traits, then fed the merged result into a compact two-stage prediction module. Since some labels appeared far more than others, extra weight was given during training to balance influence across classes. This version ran through five full cycles using identical settings for step size and group count as its simpler counterpart.

This method reached 65.51% accuracy on test data, along with a macro F1 score of 0.645 - topping all models tested. Notably, the fake news detection rate climbed from 0.467 when using only text to 0.584; previously, it had stood at 0.522 under the strongest baseline. Full results appear in Table 2.

Full Model Comparison with DistilBERT Configurations

Model	Accuracy	Fake Precision	Fake Recall	Fake F1	Real F1	Macro F1
Logistic Reg.	62.59%	0.590	0.468	0.522	0.693	0.607
Naive Bayes	62.35%	0.592	0.443	0.507	0.696	0.601
Linear SVM	60.38%	0.548	0.530	0.539	0.653	0.596
Random Forest	62.19%	0.595	0.420	0.492	0.699	0.596
Distilbert (text only)	62.59%	0.617	0.376	0.467	0.712	0.590
Distilbert (Metadata)	65.51%	0.617	0.555	0.584	0.705	0.645

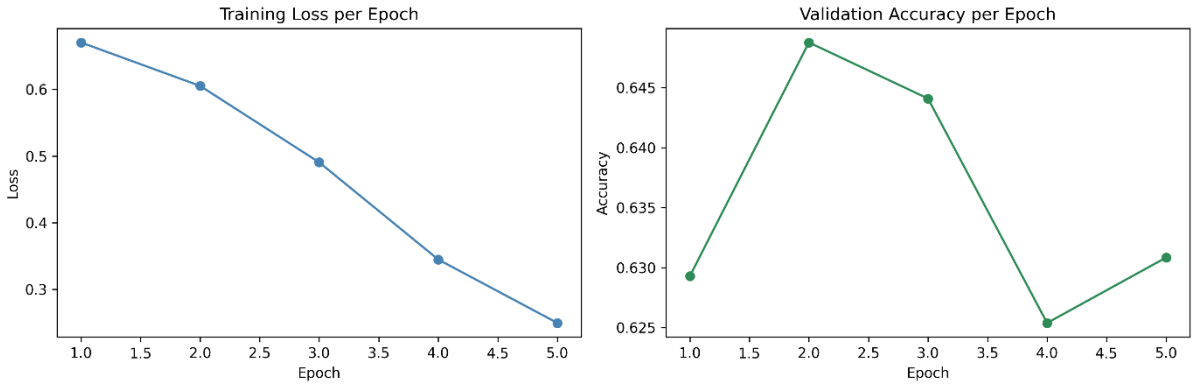


Figure 10: DistilBERT + Metadata Training loss and validation accuracy per epoch

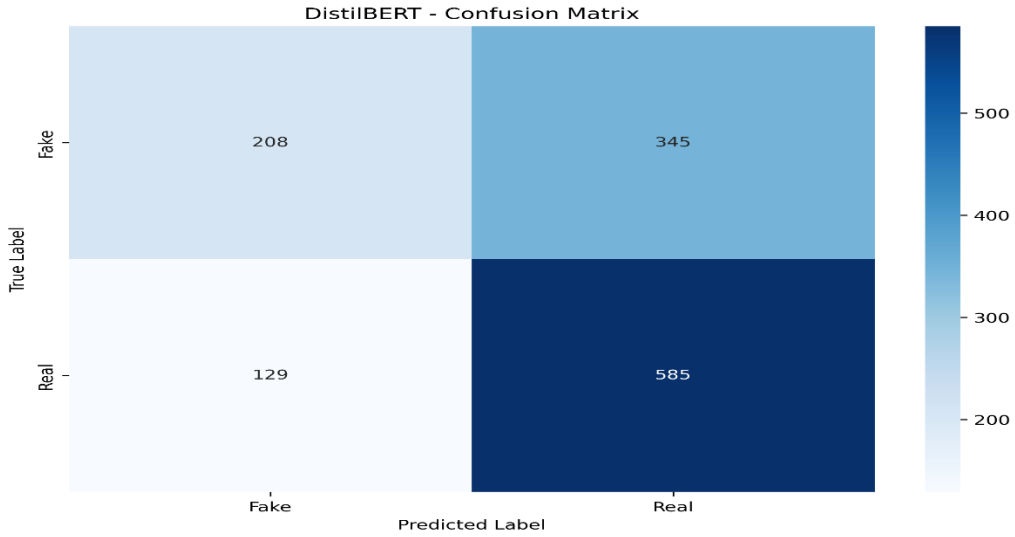


Figure 11: DistilBERT + Metadata Confusion matrix

Starting with metadata boosts performance, showing past speaker behavior matters when spotting false claims in the LIAR data. When someone often earns a "pants-on-fire" rating, their statements tend to be untrue - even if phrased convincingly - something the system picks up well. Far from being just noise, these patterns hint that detectors work better when they blend source traits with what's actually said, provided those details exist. The enhanced performance may be attributed to the design choice of the architecture to combine the DistilBERT CLS token representation with the normalised metadata features in a direct manner prior to the classification head. The CLS token acts as an abstract summary of the whole sequence of inputs, which carries the meaning of the text, whereas metadata vector adds signal of source-level credibility that is completely independent of the text itself. The simplicity and reduced computational cost of concatenation

compared to more specialized fusion techniques like cross-attention, as well as the low-dimensional nature of the metadata features and the fact that no additional contextual processing is needed to interpret them, led to its use as opposed to more complex fusion techniques.

1.5.6 Explainability Analysis

Using LIME alongside SHAP helped uncover how predictions form within Logistic Regression and DistilBERT combined with Metadata, offering insight into differing logic paths between simpler and complex models. Though based on word presence alone, the Logistic Regression model links terms like "Walker", "Scott", and "scheme" to Fake classifications when analyzing a fabricated instance - suggesting it leans heavily on names and charged expressions. A genuine sample shifts toward Real due to words including "million", "federal", and "has", which echo phrasing common in official political discourse. Instead of focusing on single cases, SHAP aggregates influence across instances; its visualization reveals "than" and "percent" stand out repeatedly among 100 examples tested. These frequent markers align closely with earlier observations about numerical patterns shaping key decisions in the logistic setup.

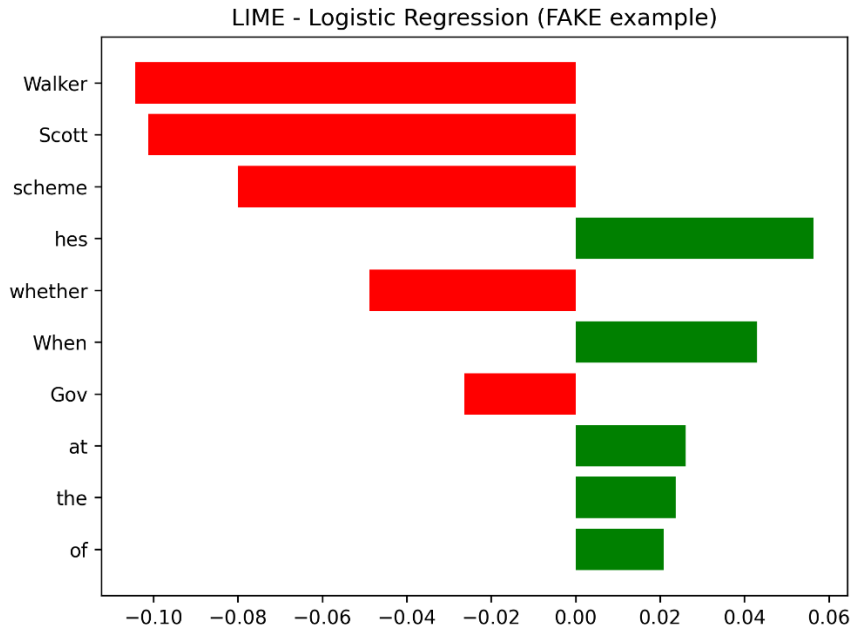


Figure 12: LIME explanation Logistic Regression (Fake example)

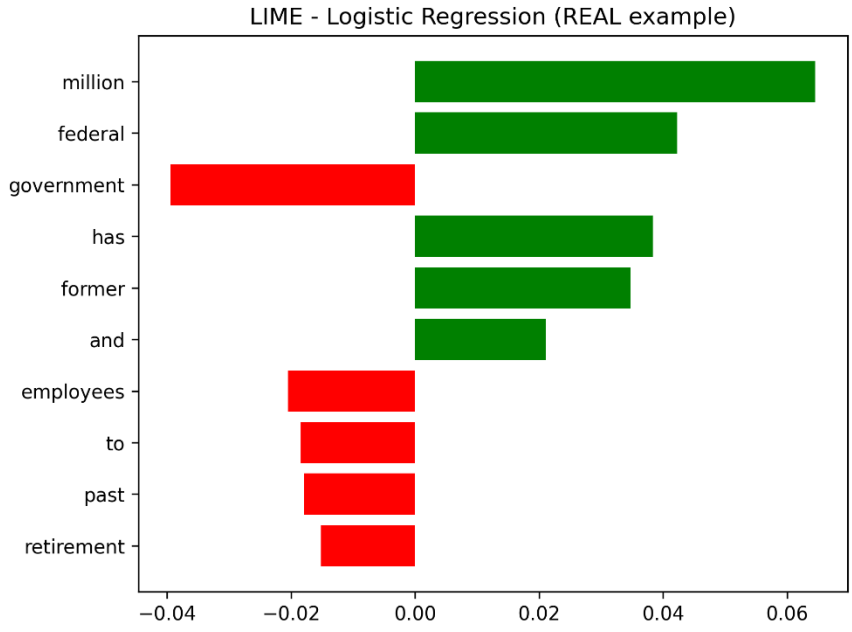


Figure 13:LIME explanation Logistic Regression (Real example)

With DistilBERT plus metadata, LIME shows shifts in reasoning across prediction classes. Instead of relying on isolated terms, it picks up phrases like "benefits", "federal", and "Over" when labeling claims as true - while false ones pull attention toward "former", "years", and "government". This points to an understanding of timing cues intertwined with political phrasing, going past mere word spotting. Unlike logistic regression, where signals are more linear, these explanations expose a deeper grasp of structure and context shaped by the architecture itself.

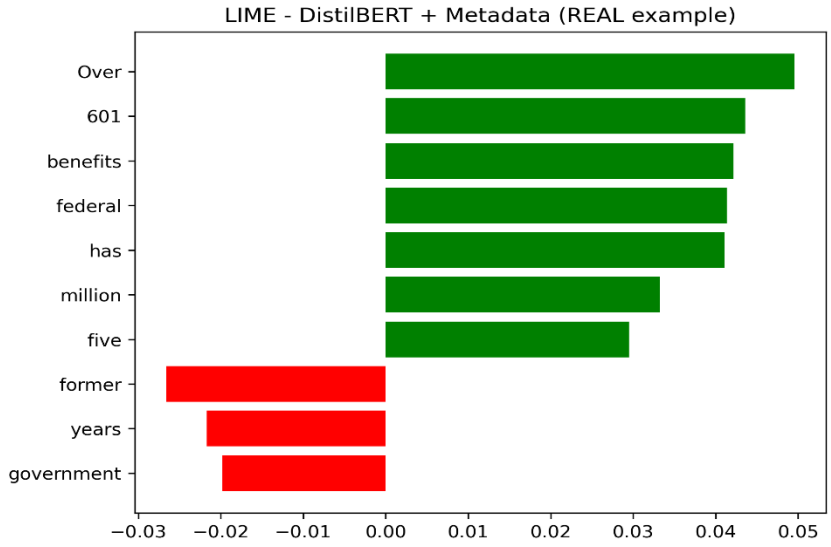


Figure 14: LIME explanation DistilBERT + Metadata (Fake example)

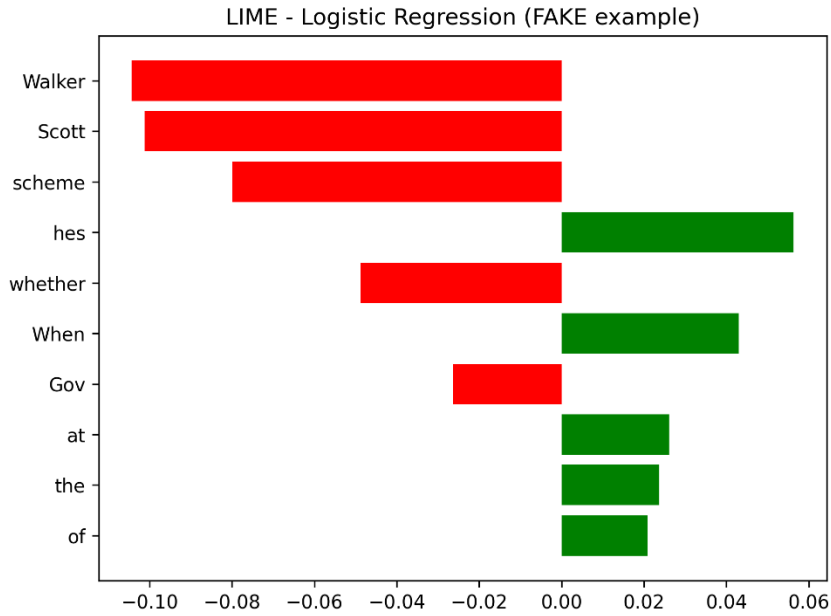


Figure 15LIME explanation DistilBERT + Metadata (Real example)

1.5.7 Interface

Through a web setup built with Gradio, the system's actual use comes into view. One picks among five available models, types in a claim, then gets an instant forecast - complete with certainty level and LIME-backed reasoning. When using DistilBERT plus metadata, extra boxes appear for party label and past speaker activity. Base setups answer fast, often under seconds; the DistilBERT variant runs just slightly longer when powered by GPU. Look to Appendix A for visual examples of how it appears during operation.

1.6 Evaluation of the Product

1.6.1 Evaluation Approach

The product created during this project is a two-part product: a collection of machine learning models to detect fake news, and a web interface based on Gradio that will allow real-time classification with explainability-output. The system was not evaluated with recruited participants through formal user evaluation as it is a research prototype and not a consumer-facing product. Rather, it is evaluated on three grounds, namely, performance benchmarking against published results on the LIAR dataset, evaluation against the objectives of the project, and critical evaluation of the strengths and weaknesses of the product.

1.6.2 Performance Against Published Benchmarks

Looking into past work shows the LIAR dataset stands out as tough, where usual transformer setups land around 59 to 65 percent correct guesses. Atop the tested approaches here sits DistilBERT paired with extra data details, hitting 65.51 percent right answers along with a balanced score - macro F1 - at 0.645, just shy of the top mark seen so far. Even though computing power was tight, restricting tests to leaner models instead of heavier ones like RoBERTa or complete BERT versions, the outcome still holds weight.

Surprisingly, DistilBERT using only text - accuracy at 62.59%, macro F1 of 0.590 - does not beat Logistic Regression, which scores the same accuracy but slightly higher at 0.607 in macro F1. This lines up with earlier reports: transformers often underperform on brief texts. Short snippets appear to limit their advantage. Take LIAR dataset - for instance - the typical claim spans just a few words, posing challenges even for advanced methods, as Wang pointed out back in 2017. When speaker details enter the model, performance climbs to 65.51%. That jump, compared to plain text alone, shows who said what carries weight. Source context adds real value; it's more than plugging in a prebuilt transformer and calling it done.

1.6.3 Assessment Against Project Aims

The project defined four aims, all of which have been addressed:

- Aim 1 — testing four traditional machine learning baselines - was completed without exception. Each model underwent training and assessment under identical data preparation and testing conditions. Documentation included detailed classification summaries, visual confusion arrays, rankings of predictive features, along with hand-labelled misclassification patterns. The process concluded having covered every required analytical step thoroughly.
- Aim 2 — testing if metadata boosts DistilBERT's results - success achieved. One setup used extra data cues, the other did not; differences showed up sharply in outcomes. Performance rose notably in accuracy and Fake F1 when metadata entered the model. Closer inspection revealed how those gains unfolded, linking input changes directly to stronger predictions.
- Aim 3 — using LIME and SHAP to enable explanation - was completed successfully. While focusing on Logistic Regression and DistilBERT combined with Metadata, each method generated clear interpretations exposing how language features influence predictions. These outputs uncovered underlying word-level trends guiding classification outcomes across models.
- Aim 4 — develop a Gradio interface — was fully met. The interface supports all five models, provides real-time predictions with confidence scores, and generates LIME explanation charts dynamically. Screenshots are provided in Appendix A.

1.6.4 Strengths of the Product

Although the system shows multiple strong points, its first advantage lies in comparing six model setups; this allows a balanced view of various methods tackling one task, offering insights beyond just highlighting top results. Because it uses LIME, explanations emerge clearly - readers spot exactly which terms shaped a given outcome instead of blindly accepting outputs. While many models act like sealed boxes, here transparency builds confidence through visibility into word-level impact. Where short texts challenge standard transformers, adding metadata to DistilBERT lifts effectiveness meaningfully. Even without coding skills,

someone can type a phrase and get both classification and intuitive visuals guiding understanding. Not every design considers ease so directly, yet simplicity remains central throughout.

1.6.5 Weaknesses of the Product

Acknowledging certain limitations is necessary here. Most notably, the metadata framework depends entirely on specific datasets - particularly PolitiFact's prior ratings of speaker trustworthiness. When applied beyond controlled settings, someone submitting an arbitrary news headline could lack any record of that source's past inaccuracies; under such conditions, metadata values drop to zero. Without those inputs, the system loses its edge. As a result, strong performance within the LIAR dataset does not guarantee similar outcomes elsewhere.

Achieving just 65.51% accuracy may seem reasonable compared to the LIAR dataset standards, yet errors occur in nearly every third prediction - raising concerns if used independently. Without active review by people, such error rates limit real-world reliability. Rather than working alone, its role fits better alongside human judgment. One out of three wrong calls show why support - not replacement - is the right frame.

Running SHAP in real time demands high computation, so it was left out despite its potential benefits. The system uses LIME alone because of this limitation - SHAP's broader insights are missing as a result.

Ultimately, there was no structured assessment involving users. Though the system operates correctly and produces understandable LIME results, insights from real-world journalists or verification professionals would still be needed to confirm if these explanations hold practical value. Despite working as designed, usefulness in daily workflows remains untested without input from those actually doing the work.

1.6.6 Informal User Evaluation

To complement the quantitative assessment, informal user assessment was done on four respondents. Participants were undergraduate students with no prior experience of fake news detection systems. Each subject was presented with a screen recording of the interface at work, where the statements of news were categorized using a baseline model and the DistilBERT + Metadata configuration, with the LIME explanation output. After the demonstration, the participants were posed five questions related to the understanding of the prediction outcome, usefulness of the LIME explanation, ease of use, system trustworthiness and recommendations on improvement.

The prediction outcome, with the Fake or Real label and a confidence score shown, was clear and understandable to all four participants. The answers to the LIME explanation chart were more conflicted. Two respondents were useful in gaining an insight into how the system decides, with one saying that it takes a little concentration to comprehend. One respondent said that the LIME chart was not helpful, and one said that it was acceptable but messy. This implies that although LIME offers users interpretable output as long as they are willing to work with it, the existing visualisation can pose a challenge to less technical users.

There was a consistent rating of the interface as easy to use by all participants with three of them rating it very easy. In terms of trustworthiness, the participants rated moderately, responses indicated that the system might be used as a starting point to check facts and not a final source, which suits the desired application of the system as a decision-support tool and not a standalone classifier.

The most uniform feedback had to do with visual presentation - four out of the participants recommended to make the interface more attractive and to make the LIME charts more readable. The results point to one obvious way of the further development: refining the visual presentation of the explanation output in order to make it more easily available to non-technical users.

1.7 Evaluation of the Project

1.7.1 Review of Aims and Progress

Early on, the work set four goals: testing traditional machine learning methods, exploring transformer models both with and without extra data inputs, using tools to interpret model decisions, plus building a functional

demo system. Each target was completed by the due date. Overall, the effort stayed close to the initial plan - except for one key change. A version of DistilBERT boosted with metadata was added after early tests showed plain text-based DistilBERT did no better than basic Logistic Regression. Adjusting course like this fit's standard scientific inquiry, showing how checking progress step-by-step helps guide smarter next moves.

1.7.2 What Went Well

What stood out most was how the experiment flow stayed tightly organised. Because preprocessing, turning data into vectors, and testing followed the same steps every time, one model could be measured clearly against another - this made conclusions more trustworthy. Using distinct Jupyter notebooks for different parts of the process helped keep things repeatable; going back to fix or check a single piece did not interfere with the rest. A clear structure turned complexity into something manageable.

Surprisingly, digging into metadata turned out to be worthwhile. At first, everything revolved around sorting texts without extra cues; however, when basic DistilBERT struggled, doubts emerged about relying solely on words. A shift began once past speaker reliability was seen not just as background detail, but as useful signal. Building a system that wove this aspect into its core logic became the hardest yet most satisfying piece of work.

Surprisingly clear results came from combining LIME with SHAP. While one method highlighted specific word contributions, the other reinforced patterns seen during manual review of mistakes. Together, they supported each piece of evidence without contradiction. From start to finish, interpretations stayed close to actual language use. Each approach echoed what the numbers had already suggested. Coherence emerged naturally between stats and real examples.

1.7.3 Challenges and How They Were Addressed

Unexpected outcomes shaped much of the work. Notably, the first setup using DistilBERT did not meet expectations. Despite five rounds of tuning, its precision matched only that of Logistic Regression, while showing weaker average F1 scores - surprising, considering how well similar transformers usually perform. Instead of moving on, attention shifted toward testing whether metadata could explain the gap. That shift led to the top-performing version. What began as a setback became useful; poor results often reveal more when examined closely.

Another issue came down to limited hardware. Running DistilBERT on a GTX 1660 Super, equipped with just 6GB of VRAM, meant adjusting batch sizes and shortening sequences - otherwise, memory overflow occurred. Larger architectures like RoBERTa or standard BERT were not feasible under these conditions, even though performance might have improved. Later versions could explore broader options if cloud-based GPUs become available, perhaps through services like Google Colab Pro or institutional computing clusters.

A key hurdle involved handling how hard the LIAR dataset really is. Though one claim might be only slightly off while another is wildly false, both get stamped as equally fake under the current two-label system - flattening meaningful differences between them. Because of this merging, mistakes during testing often trace back to mixed-up categories. Such misclassification stems directly from reducing complex truths into just two groups - a constraint baked into any simplified setup unless more labels are introduced.

1.7.4 Skills and Learning Developed

This was a project that enhanced technical and academic skills. Technically, there was experience with fine-tuning transformer models with the HuggingFace Transformers library, designing custom neural network architectures in PyTorch that fuse heterogeneous input types, and using post-hoc explainability methods with both classical and deep learning models. To construct the metadata-enhanced DistilBERT model, it was necessary to learn how to combine dense vectors representations with structured numeric features, which is applied in the production recommendation and classification systems. Implementing a working demonstration using Gradio provided experience working with machine learning productionisation.

In the academic sphere, the project has brought the skill to analyze the outcomes of experiments critically rather than have issues with it in its surface. The choice to research the metadata hypothesis after DistilBERT

failed its initial test, instead of merely reporting the negative finding, is indicative of the type of analytical reasoning that can be considered the difference between strong research and mechanical implementation. The ability to contextualise findings in the published literature, know where results coincide and where they differ with existing literature, and make valid conclusions based on quantitative data are skills that can be transferred to practice in both research and industry settings.

The project also emphasised the role of experimental rigour - keeping the same preprocessing pipelines on all models, optimising the vectoriser only to training data, and macro F1 instead of the accuracy as the main measure were all thoughtful methodological decisions that enhanced the soundness of the comparative analysis.

1.7.5 What Would Be Done Differently

Earlier attention to metadata analysis might change how future versions unfold, shifting it forward instead of treating it like a fix after weak results show up. Since speaker details exist in the LIAR set right away, weaving them into the transformer architecture at launch could open space for deeper adjustments later, maybe lifting outcomes. Testing explanations through structured sessions with real users - say, reporters or verification specialists - may ground claims about usefulness in actual experience, adding weight to judgments about value.

1.8 Conclusions

This work aimed at building a tool to spot false information through machine learning, relying on transformer architectures and tested with the LIAR collection. Although focused on accuracy, it first introduced standard algorithms as reference points - then measured their performance alongside newer approaches. Instead of stopping at predictions, one part explored model behavior using LIME and SHAP for transparency. With DistilBERT, tests ran twice: once reading only words, another time including background details like source and context. A live demo followed, made interactive via Gradio, letting users submit claims and get instant feedback. Every original target was ultimately reached by combining these steps.

What stands out most from the study is how architectural complexity by itself does not guarantee better results on the LIAR data. Without longer texts, even advanced models like DistilBERT struggle - its text-only version couldn't beat basic Logistic Regression, suggesting limited room for attention layers to help. Instead of depending only on words, adding details about speakers made a real difference - using past behavior and political ties along with language features lifted performance noticeably. That combined approach reached 65.51% accuracy and a macro F1 score of 0.645, topping all five other setups tested. When building systems to spot false claims, one takeaway becomes clear - if background facts about sources exist, pairing them with text beats treating language in isolation. Not every gain comes from deeper networks; sometimes context outside the sentence matters more.

Another major result involves how similar the mistakes are between different systems. Looking closely at errors showed that unclear meanings and numbers were behind most incorrect judgments, no matter which of the five systems was used - indicating the real issue lies more in how tough the data is than in what system processes it. Phrases with uncertain number estimates or just minimal subject hints kept causing problems repeatedly, highlighting a core obstacle in checking brief political claims that switching models won't fix by itself.

Looking closely at how models decide, it became clear that Logistic Regression and DistilBERT combined with metadata rely on understandable linguistic cues. Words like "than", "percent", or "more" appear repeatedly when the systems flag something as false - highlighted through LIME's local insights. True claims, instead, tend to carry markers of official sources or verified facts. These tendencies aren't limited to isolated cases; SHAP showed they hold across large samples. Trust improves when we can see why a decision was made, especially since many current detectors give answers without any reasoning behind them.

Future Work

One path forward involves testing bigger transformer models like RoBERTa or BERT-large instead of DistilBERT, since they often score better on classification tasks when enough computing power is available. Though smaller, current setups might benefit from scale. Another option emerges through structured feedback: involving journalists or fact-checking professionals in evaluating how well LIME-generated explanations support real-world decision making. Because human judgment plays a role here, their insights matter. While the top model relies heavily on metadata, that limitation might fade if linked to an up-to-date source tracking speaker credibility across cases beyond just those found in the LIAR collection. As new data enters, trust signals update. Since false claims now travel using photos, clips, and sound as much as words, adding these forms into analysis could reflect today's communication patterns more accurately. Even text alone does not capture everything anymore.

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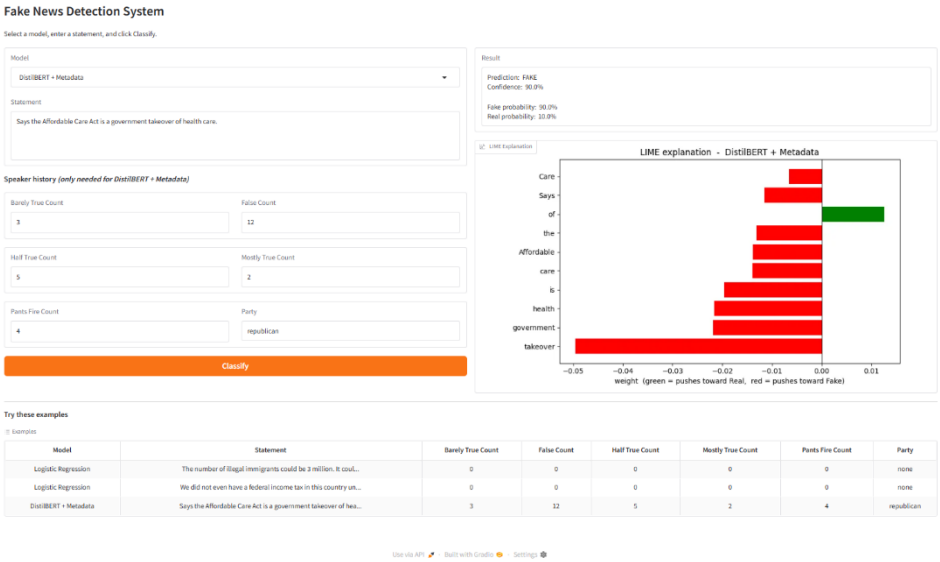
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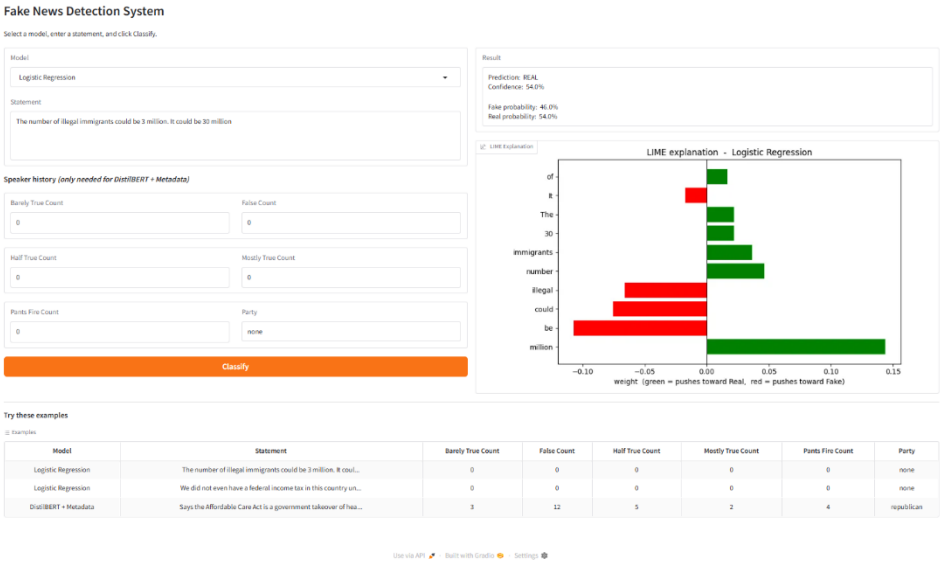
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Singh, A., Shukla, H., Sharma, P., Srivastava, S., & Mishra, P. K. (2026). A hybrid machine learning approach with explainable AI for predicting fake news on social media platforms. *Journal of Supercomputing*.

Wang, W. Y. (2017). "Liar, liar pants on fire": A new benchmark dataset for fake news detection. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (pp. 422–426).

Appendix A: Interface Screenshots





Appendix B: Ethics Form

PROJECT ETHICAL REVIEW FORM

Applicable for all research, masters and undergraduate projects

Project Title:	Fake News Detection Using Machine Learning and Transformer-Based Models
Student:	Azan Gaziani
Course/Programme:	BSc Computer Science with AI
Department:	School of Computing and Engineering
Supervisor:	Ilias Tachmazidis
Project Start Date:	December 2025

ETHICAL REVIEW CHECKLIST

	Yes	No
1. Are there problems with any participant's right to remain anonymous?	☐	X
2. Could a conflict of interest arise between a collaborating partner or funding source and the potential outcomes of the research, e.g. due to the need for confidentiality?	☐	X
3. Will financial inducements be offered?	☐	X
4. Will deception of participants be necessary during the research?	☐	X
5. Does the research involve experimentation on any of the following?		
(i) animals?	☐	X
(ii) animal tissues?	☐	X
(iii) human tissues (including blood, fluid, skin, cell lines)?	☐	X
6. Does the research involve participants who may be particularly vulnerable, e.g. children or adults with severe learning disabilities?	☐	X
7. Could the research induce psychological stress or anxiety for the participants beyond that encountered in normal life?	☐	X
8. Is it likely that the research will put any of the following at risk:		
(i) living creatures?	☐	X
(ii) stakeholders (disregarding health and safety, which is covered by Q9)?	☐	X
(iii) the environment?	☐	X
(iv) the economy?	☐	X
9. Having completed a health and safety risk assessment form and taken all reasonable practicable steps to minimise risk from the hazards identified, are the residual risks acceptable (Please attach a risk assessment form – available at the end of this document)	X	☐

STATEMENT OF ETHICAL ISSUES AND ACTIONS

If the answer to any of the questions above is yes, or there are any other ethical issues that arise that are not covered by the checklist, then please give a summary of the ethical issues and the action that will be taken to address these in the box below. If you believe there to be no ethical issues, please enter "NONE".

None

STATEMENT BY THE STUDENT

I believe that the information I have given in this form on ethical issues is correct.

Signature: Azan Gaziani Date: 19/04/2026

AFFIRMATION BY THE SUPERVISOR

I have read this Ethical Review Checklist and I can confirm that, to the best of my understanding, the information presented by the student is correct and appropriate to allow an informed judgement on whether further ethical approval is required.

Signature: Ilias Tachmazidis Date: 20/04/2026

SUPERVISOR RECOMMENDATION ON THE PROJECT'S ETHICAL STATUS

Having satisfied myself of the accuracy of the project ethical statement, I believe that the appropriate action is:

The project proceeds in its present form	
The project proposal needs further assessment by an Ethical Review Panel. The Supervisor will pass the form to the Ethical Review Panel Leader for consideration.	

RETENTION OF THIS FORM

- The Supervisor must retain a copy of this form until the project report/dissertation is produced.
- The student must include a copy of the form as an appendix in the report/dissertation.

OUTCOME OF THE ETHICAL REVIEW PANEL PROCESS, <u>WHERE REQUIRED</u>

Tick One

1. Approved. The ethical issues have been adequately addressed and the project may commence. ☐

2. Approved subject to minor amendments. The required amendments are stated in the box below. The project may proceed once the form has been amended in line with the requirements and signed by the Supervisor in the box imediately below to confirm this. ☐

I confirm, as Supervisor, that the amendments required have been made:

Signature: _____ Date: _____

3. Resubmit. The areas requiring further action are stated in the box below. The project may not proceed until the form has been resubmitted and approved. ☐

4. Reject. The reasons why it will not be possible to address the ethical issues adequately are stated in the box below. ☐

For any of the outcomes 2, 3 or 4 above, please provide a statement in the box below.

AFFIRMATION BY THE REVIEW PANEL LEADER

I approve the decision reached above by the review panel members:

Signature: _____ Date: _____

THE UNIVERSITY OF HUDDERSFIELD: RISK ANALYSIS & MANAGEMENT

ACTIVITY:			Name:	
LOCATION:			Date:	Review Date:
Hazard(s) Identified	Details of Risk(s)	People at Risk	Risk management measures	Other comments
Prolonged screen use	Eye strain and fatigue during extended coding and writing sessions	Student	Regular breaks taken, screen brightness adjusted, work sessions limited to manageable durations	Risk level is low and acceptable

Appendix C


FAKE NEWS DETECTION
Using Machine Learning and Transformer-Based Models
with Speaker Metadata and Explainable AI
Azan Gaziani U2388422
BSc Computer Science with Artificial Intelligence
University of Huddersfield
Supervisor: Ilias Tachmazidis
Final Year Project — May 2026


Agenda
1
Introduction
Problem, motivation, and aims
2
Dataset and Methodology
LIAR, design decisions, preprocessing
3
Baseline Models
Implementation, results, error analysis
4
DistilBERT Models
Text-only and metadata-enhanced configurations
5
Explainability
LIME and SHAP interpretation
6
Demo, Evaluation, Conclusions
Live system and findings

The Problem

Fake news — deliberately fabricated information designed to deceive — has real-world consequences:

> **2016 US Election**
>
> Fabricated content spread faster than real news on social media

> **Southport, July 2024**
>
> False reports of attacker's identity led to UK civil disturbances

> **COVID-19 Pandemic**
>
> False vaccine claims contributed to preventable deaths

This is not just an information quality problem — it is a public safety issue.

Scale of the Problem

6×	68%	1M+
faster	*of Americans*	*messages*
Misinformation spreads vs truth on social media	Now obtain news from social media platforms	Disinformation campaigns from 2,000+ accounts

Manual fact-checking cannot match this volume — automated detection is necessary.

Existing Solutions and Gaps

Manual Platforms

- **PolitiFact, Snopes, Full Fact**
- Reliable but slow
- False content typically spreads before fact-checks publish
- Cannot scale to modern volumes

AI-Based Tools

- **Intel FakeCatcher (deepfakes)**
- **Bot Sentinel (Twitter accounts)**
- Modality or platform specific
- Most lack explainability

Gap addressed by this project: combine accuracy, transparency, and accessibility in one system

Project Aims

1 **Implement and Evaluate Baselines**

Four classical ML models on the LIAR dataset

2 **Investigate Metadata Enhancement**

Compare DistilBERT text-only vs DistilBERT + speaker metadata

3 **Apply Explainability Methods**

LIME and SHAP across classical and transformer models

4 **Build Interactive Interface**

Gradio-based system with real-time explanation

Dataset: LIAR

12,836	19	6→2
labelled statements	average words per claim	labels collapsed to binary

Why LIAR?

- Sourced from PolitiFact.com
- Labels: pants-fire, false, barely-true → Fake (0); half-true, mostly-true, true → Real (1)
- Includes speaker metadata: party affiliation and historical lie counts
- Genuinely difficult — even SOTA transformers achieve only 59-65% accuracy

Methodology Overview



CRISP-DM iterative process — each stage informs the next

Key Design Decisions

Every choice was deliberate, not default:

LIAR over ISOT

ISOT models 'cheat' by detecting publisher names like 'Reuters'

DistilBERT over BERT/RoBERTa

97% performance at 40% the size — fits 6GB GPU

No stopword removal

Bigrams like 'more than' carry discriminative signal

Concatenation fusion

Simpler than cross-attention; suits low-dim metadata

TF-IDF fitted on training only

Prevents data leakage into test set

Macro F1 over accuracy

Addresses 553 Fake vs 714 Real class imbalance

These decisions ensure fair comparison and prevent the 'black box' anti-pattern

Preprocessing & Feature Extraction

Text Cleaning Pipeline

Custom clean_text() function using Python regex:

- Lowercase conversion via .lower()
- URL removal via re.sub(http|www patterns)
- Strip special characters (regex)
- Whitespace normalisation

TF-IDF Configuration

scikit-learn TfidfVectorizer parameters:

- max_features = 5000
- ngram_range = (1, 2) — unigrams + bigrams
- min_df = 2, max_df = 0.8
- fit_transform on train, transform on test

DistilBERT bypasses TF-IDF — uses HuggingFace tokeniser (max_length=128) on raw text

Baseline Models

Logistic Regression
Linear classifier, sigmoid output
max_iter=1000

Multinomial Naive Bayes
Probabilistic, feature independence
alpha=1.0

Linear SVM
Maximum-margin separator
C=1.0, max_iter=2000

Random Forest
Ensemble of decision trees
n_estimators=200

All models share identical preprocessing, TF-IDF features, and evaluation pipeline

Baseline Results

Model	Accuracy	Fake F1	Real F1	Macro F1
Logistic Regression	**62.59%**	0.522	0.693	**0.607**
Naive Bayes	62.35%	0.507	0.696	0.601
Linear SVM	60.38%	0.539	0.653	0.596
Random Forest	62.19%	0.492	0.699	0.596

Key Observations

- Logistic Regression leads on accuracy AND macro F1 — strong baseline on TF-IDF
- Real F1 ≫ Fake F1 across all models (0.15–0.21 gap) → class bias signal
- Random Forest needs 200 trees but matches Naive Bayes — diminishing returns

Error Analysis (Manual)

Process — going beyond automated metrics:

- Sampled 20 misclassified examples per model (100 total across 5 models)
- Personally read each statement and assigned to one of 5 categories
- Computed category distribution per model and visualised as pie charts

Ambiguous	Numerical	Mixed	Opinion	Other

Key Insight

Across all 5 models: **50–60% of errors are 'Ambiguous'** statements lacking enough context. Same examples failed in every model.

→ Limitation is the dataset, not the model. Motivates metadata exploration.

DistilBERT — Text-Only Configuration

Implementation:

- HuggingFace AutoModelForSequenceClassification (DistilBERT base, uncased)
- Tokeniser: max_length=128 (sufficient for 19-word avg statements)
- Training: 5 epochs, lr=2e-5, batch_size=16, AdamW optimiser
- Class weighting in CrossEntropyLoss to handle imbalance

Result: 62.59% accuracy, 0.590 macro F1

Identical accuracy to Logistic Regression — but **lower macro F1** due to weak Fake recall.

Interpretation: at avg 19 words, there isn't enough context for DistilBERT's bidirectional attention to find signal that TF-IDF misses.

DistilBERT + Metadata — Custom Architecture

Custom PyTorch class — extending DistilBERT with structured features



Why this design?

- Concatenation chosen over cross-attention (low-dim metadata, no extra context needed)
- [CLS] token captures full sentence semantics from DistilBERT
- Metadata = 5 lie-history counts + party affiliation (one-hot)

DistilBERT + Metadata — Results

65.51%	0.645	0.584
Accuracy	**Macro F1**	**Fake F1**
Best of all 6 configurations	*+0.038 vs LR baseline*	*+0.117 vs text-only DistilBERT*

Why does it work?

- Speaker credibility history is genuine signal — orthogonal to the text content
- A speaker with many 'pants-fire' ratings tends to make false claims even if phrased convincingly
- Fake recall jumps from 0.376 → 0.555 — the model now catches more genuine fake content

Full Model Comparison

Model	Accuracy	Fake F1	Real F1	Macro F1
Logistic Regression	62.59%	0.522	0.693	0.607
Naive Bayes	62.35%	0.507	0.696	0.601
Linear SVM	60.38%	0.539	0.653	0.596
Random Forest	62.19%	0.492	0.699	0.596
DistilBERT (text only)	62.59%	0.467	0.712	0.590
DistilBERT + Metadata	**65.51%**	**0.584**	**0.705**	**0.645**

Key takeaway: architectural complexity alone doesn't help. Speaker context is the differentiator.

Explainability — LIME

Local Interpretable Model-agnostic Explanations

How it works:

- Generates many perturbations of the input by removing words randomly
- Runs each perturbation through the model and observes prediction changes
- Builds a local linear approximation showing word-level influence
- Per-prediction explanation — fast and interpretable

Applied to:

Logistic Regression

Top words: 'Walker', 'Scott', 'scheme' → Fake; 'million', 'federal' → Real

DistilBERT + Metadata

Picks up phrases like 'benefits', 'over' (Real); 'former', 'years' (Fake)

Explainability — SHAP

SHapley Additive exPlanations

How it works:

- Calculates each feature's contribution by testing it across many combinations
- Provides both global (across many predictions) and local (single prediction) views
- Captures non-linear effects that LIME's linear approximation may miss

Applied to Logistic Regression

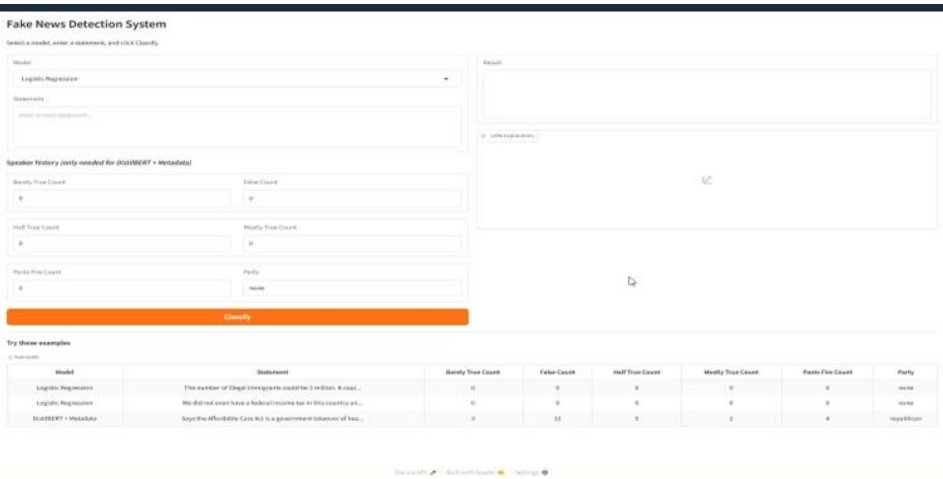
- 100 background samples + 100 test samples (LinearExplainer)
- Globally most-impactful tokens: 'than', 'percent', 'and' → Fake signals
- SHAP not applied to DistilBERT — too compute-intensive on transformers

Interactive Interface

Built with Gradio — Python framework for ML demos

Features:

- Dropdown selector for all 5 trained models
- Conditional metadata fields (appear when DistilBERT + Metadata selected)
- Real-time prediction with confidence score for both classes
- Dynamic LIME explanation chart generated per prediction
- GPU-accelerated inference — sub-second baseline, ~1s for DistilBERT



Walking through:

1. Statement input + model selection
2. Prediction with confidence score
3. LIME explanation chart
4. Comparison: LR vs DistilBERT + Metadata

Evaluation & Limitations

Strengths

- Six configurations directly comparable
- Macro F1 of 0.645 — competitive on LIAR
- Manual error analysis beyond metrics
- Explainability through LIME + SHAP
- Working interactive demo

Limitations

- Metadata only available for LIAR speakers
- 65.51% = 1 in 3 predictions wrong
- SHAP not feasible on DistilBERT (compute)
- No professional user evaluation conducted
- Hardware constrained to DistilBERT-size

Conclusions

Architecture alone isn't enough

Text-only DistilBERT did not beat Logistic Regression on LIAR — short statements limit attention's advantage

Speaker context matters

Adding metadata raised accuracy by ~3 points and Fake F1 by ~12 points — context outside the sentence is key

Errors are dataset-driven

50–60% of errors across all models share the same 'ambiguous' nature — limitation lies in the data, not the model

Explainability builds trust

LIME and SHAP revealed consistent linguistic cues — comparison terms drive Fake; named entities anchor Real

Future Work

1. **Larger transformer architectures**
   Test RoBERTa-large or BERT-large with cloud GPU resources
2. **Real speaker credibility lookup**
   Replace manual metadata with live PolitiFact API integration
3. **Multimodal extension**
   Process images and video alongside text (modern misinformation)
4. **Professional user study**
   Recruit journalists and fact-checkers to validate explanation quality

Key References

- Wang, W. Y. (2017). 'Liar, liar pants on fire': A new benchmark dataset for fake news detection. ACL.
- Sanh, V. et al. (2019). DistilBERT, a distilled version of BERT. NeurIPS Workshop.
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- Ribeiro, M. T. et al. (2016). 'Why Should I Trust You?': Explaining the predictions of any classifier. KDD.
- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. NeurIPS.

Full reference list (17 sources) provided in the report (APA 7).

Thank you

Questions?

Azan Gaziani | University of Huddersfield | May 2026